

Practical :- Fraud

Detection

STEP :- 1

(Q1) Why is Accuracy a Terrible metric for fraud Detection?

→ Credit card fraud Detection is a highly imbalanced classification problem because fraudulent transactions makeup only about 0.17% of all transactions, while 99.83% are legitimate. In such cases, accuracy alone does not reflect how well a model detects fraud.

eg :- If a model predicts every transaction as "Not fraud", it will correctly classify all legitimate transaction but fail to detect any fraudulent

eg :- 100,000 transactions

legitimate = 99,830

fraudulent = 170

Accuracy :-

$$\begin{aligned}\text{Accuracy} &= (TP + TN) / \text{Total} \\ &= (0 + 99,830) / 100,000 \\ &= 99.83\%\end{aligned}$$

Although the accuracy is extremely high the model completely fails to detect fraud.

Precision :-

$$\begin{aligned}\text{Precision} &= TP / (TP + FP) \\ &= 0 / (0 + 0) \\ &= 0\end{aligned}$$

Recall :-

$$\begin{aligned}\text{Recall} &= TP / (TP + FN) \\ &= 0 / (0 + 170) \\ &= 0\end{aligned}$$

Shows that accuracy is misleading in fraud detection because it ignores the model's ability to identify fraudulent transactions.

(Q2)

What is the Precision = Recall tradeoff?

→ Precision and Recall are the two most important evaluation metrics for fraud detection

Precision : measures how many transactions predicted as fraud are actually fraudulent.

formula : $\text{Precision} = TP / (TP + FP)$

High precision means fewer false alarms.

Recall : measures how many actual fraudulent transactions are correctly detected.

formula : $\text{Recall} = TP / (TP + FN)$

High Recall means fewer fraud cases are missed.

→ confusion matrix

Actual / Predicted	legitimate	fraud
legitimate	TN	FP
fraud	FN	TP

→ effect of lowering the Threshold

suppose the fraud classification threshold is reduced from 0.50 to 0.20

decreases :-

- more transactions are classified as fraud
- True Positives increase.
- False Positives also increase
- Recall increases because more fraud cases are detected
- Precision usually decreases because more legitimate transactions are incorrectly flagged.

(Q3) Explain 3 strategies to handle extreme class imbalance :-

- (a) Random Undersampling
- (b) SMOTE oversampling
- (c) class-weight = 'balanced'

Write Pros and Cons of each?

→ (A) Random Undersampling :-

— Reduces the Number of majority class transactions to create a balanced dataset.

* Adv :-

- Simple to implement
- faster model training
- Reduces dataset size

* Disadv :-

- Removes valuable information
- can reduce model accuracy
- may ignore important transaction patterns

→ (B) SMOTE (Synthetic Minority OverSampling Technique)

- generates synthetic fraud samples instead of simply duplicating existing ones.

- Adv :-
- preserves majority class data
 - Improves fraud detection
 - Reduces class imbalance effectively
 - often improves Recall and F1-Score

- Dis adv :-
- can generate unrealistic samples
 - may increase overfitting
 - computationally more expensive.

→ class_weight = 'balanced' :-

- instead of changing the dataset this approach assign higher importance to minority class samples during model training.

- Adv :-
- No data loss
 - Easy to implement
 - works well for logistic Regress and Random forest.
 - does not create Synthetic data

- Disadv :-
- performance depends on algorithm.
 - may not completely solve severe imbalance.

Q4) What is the precision-Recall AUC (PR-AUC)? why is it preferred over ROC-AUC for extreme imbalance datasets?

→ precision Recall Area Under the curve (PR-AUC) measures the model's ability to balance precision and recall across different classification thresholds.

Unlike ROC-AUC, PR-AUC focuses mainly on the minority

→ Why PR-AUC is preferred:-

- more suitable for highly imbalanced datasets
- measures fraud detection performance directly.
- sensitive to improvements in precision and recall
- Better reflects real business performance than ROC-AUC.

Q5) Business impact of false positives and false negative?

→ False Positive (FP)

A legitimate transaction is incorrectly classified as fraudulent.

* Business Impact :-

- Genuine customer payment declined
- customer frustration increase
- manual investigation is required

→ eg False negative (FN)

A fraudulent transaction is incorrectly classified as legitimate.

* Business Impact :-

- fraudulent payment is approved
- financial loss occurs
- chargeback costs increase
- company reputation affected